

llm personality bias — analysis report

professor discussion · May 20, 2026

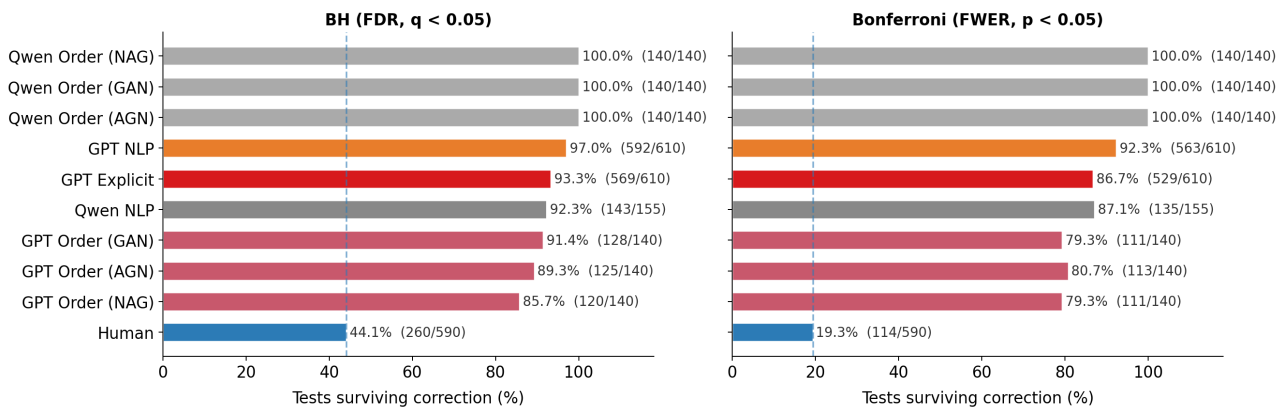
- 19,720 human ipip-50 responses as baseline (age, gender, race, country)
- gpt-4o simulated in two conditions: explicit prompting and nl (backstory) prompting
- five big-five traits: extroversion, agreeableness, conscientiousness, emotional stability, openness
- pipeline: bh/bonferroni correction → factor-level anova → explicit vs nlp comparison → alignment metrics → variance collapse
- qwen nlp excluded from primary results (n=1,200 vs ~20k for gpt) — stronger models planned

1. methodology

- human data: ipip-50 questionnaire, 1–5 likert scale, 10 items per trait, reverse-scored where needed
- explicit condition: model given demographic attributes directly and asked to fill in ipip items as that person
- nlp condition: demographic profile converted to a natural-language backstory; model responds in character
- trait score = mean of 10 scored items per trait (emotional stability = reversed neuroticism)
- subgroup stats (mean, std, n) computed for each single-factor demographic slice (e.g. gender=female)
- significance: welch t-test vs global human baseline, corrected with benjamini–hochberg ($q < 0.05$) and bonferroni
- anova: one-way per factor×source, two-way for factor pairs, full model: trait ~ gender + age + race + prompt type + model
- alignment: pearson r between human subgroup mean vector and model subgroup mean vector per factor×trait
- variance collapse: ratio of within-subgroup σ (model ÷ human) per subgroup × trait

2. multiple hypothesis testing (bh + bonferroni)

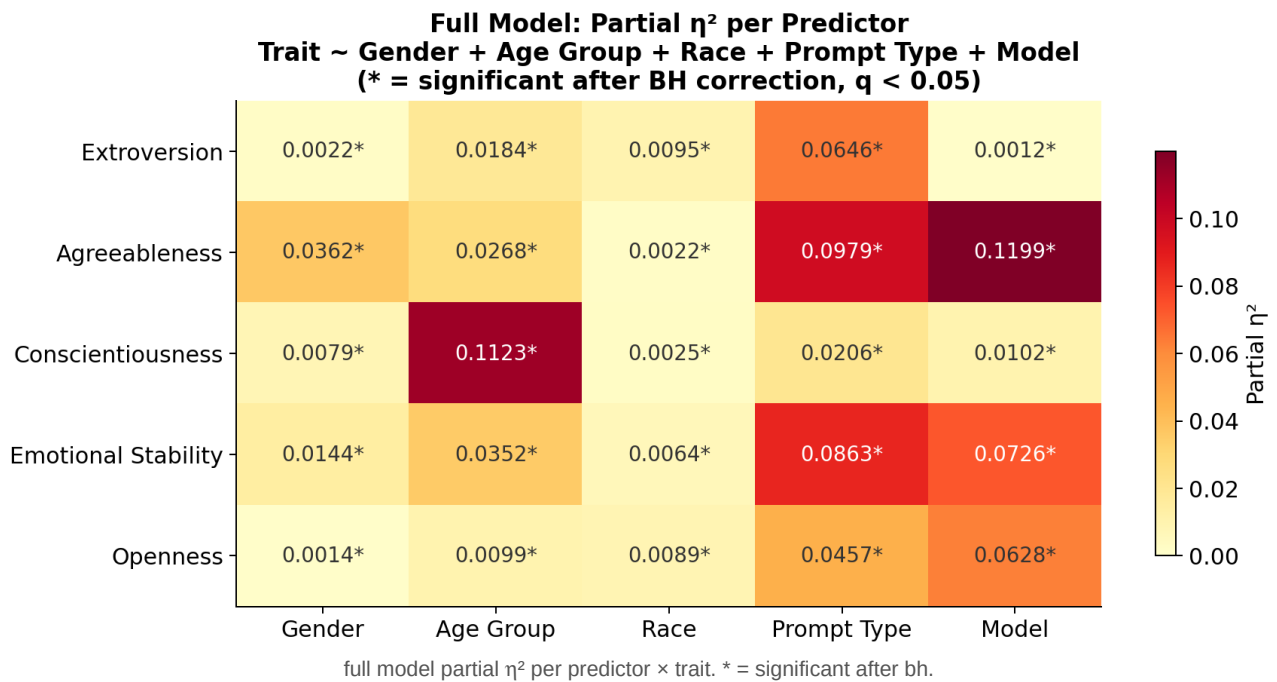
Multiple Testing Correction: Survival Rate by Source
(dashed = Human baseline)



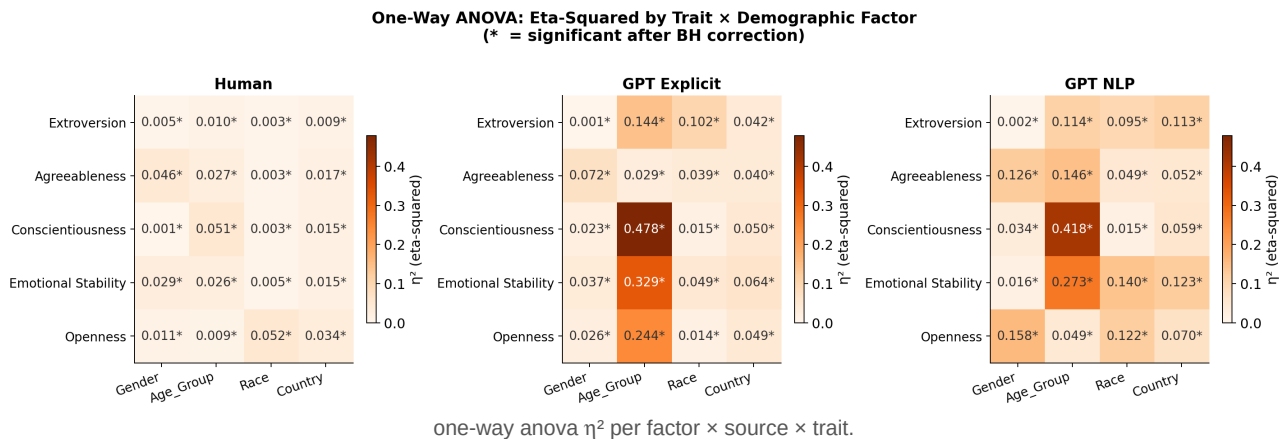
bh and bonferroni survival rates per source. dashed line = human baseline.

- human: 44% of subgroup tests survive bh correction
- gpt explicit: 93% survive — nearly all subgroup effects are statistically robust
- gpt nlp: 97% survive
- high survival rate for gpt does not mean alignment with humans — it means the model is consistently wrong in the same direction
- bonferroni is more conservative but the ordering across sources is identical

3. factor-level anova

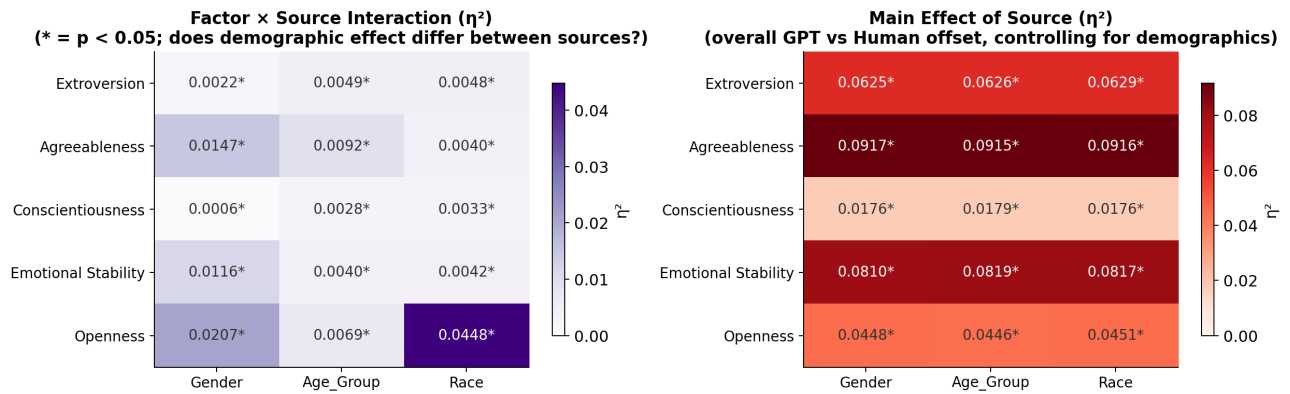


- full model: trait ~ gender + age group + race + prompt type + model
- prompt type and model are among the strongest predictors — in several traits they explain more variance than any demographic factor
- agreeableness most affected: $\eta^2=0.120$ for model, $\eta^2=0.098$ for prompt type
- conscientiousness shows the strongest age group effect ($\eta^2=0.112$) — consistent across human and gpt
- race and gender effects are smaller but significant across most traits



- age group has the most consistent effect across human, gpt explicit, and gpt nlp
- gpt nlp generally produces larger factor effects than humans for most traits
- country effect is weakest for both humans and models — noisy, limited shared subgroups

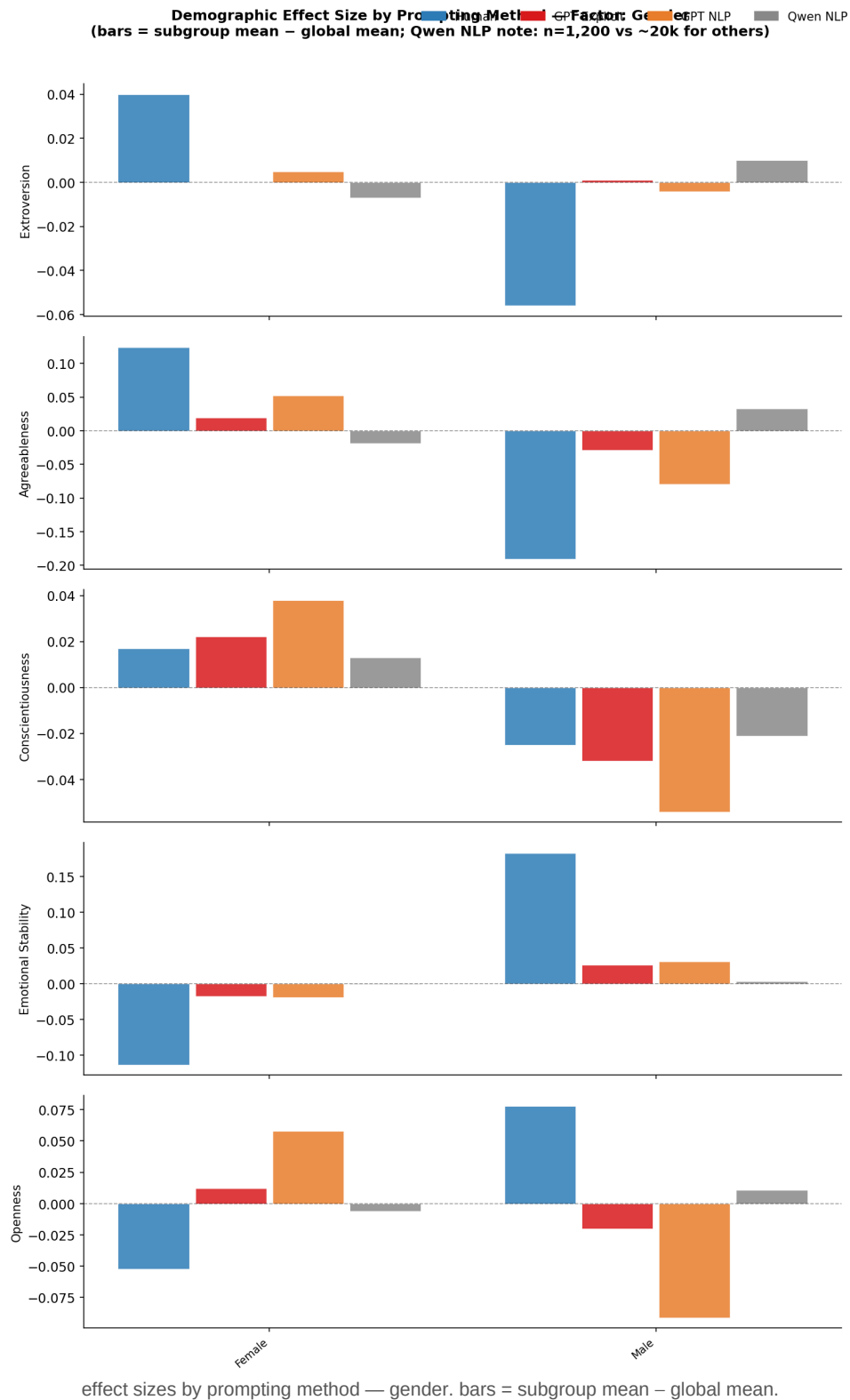
Does the Demographic Effect Differ Between Human / GPT Explicit / GPT NLP?
Two-way ANOVA: Trait ~ Factor + Source + Factor×Source



source × factor interaction: does the demographic effect differ across human/gpt explicit/gpt nlp?

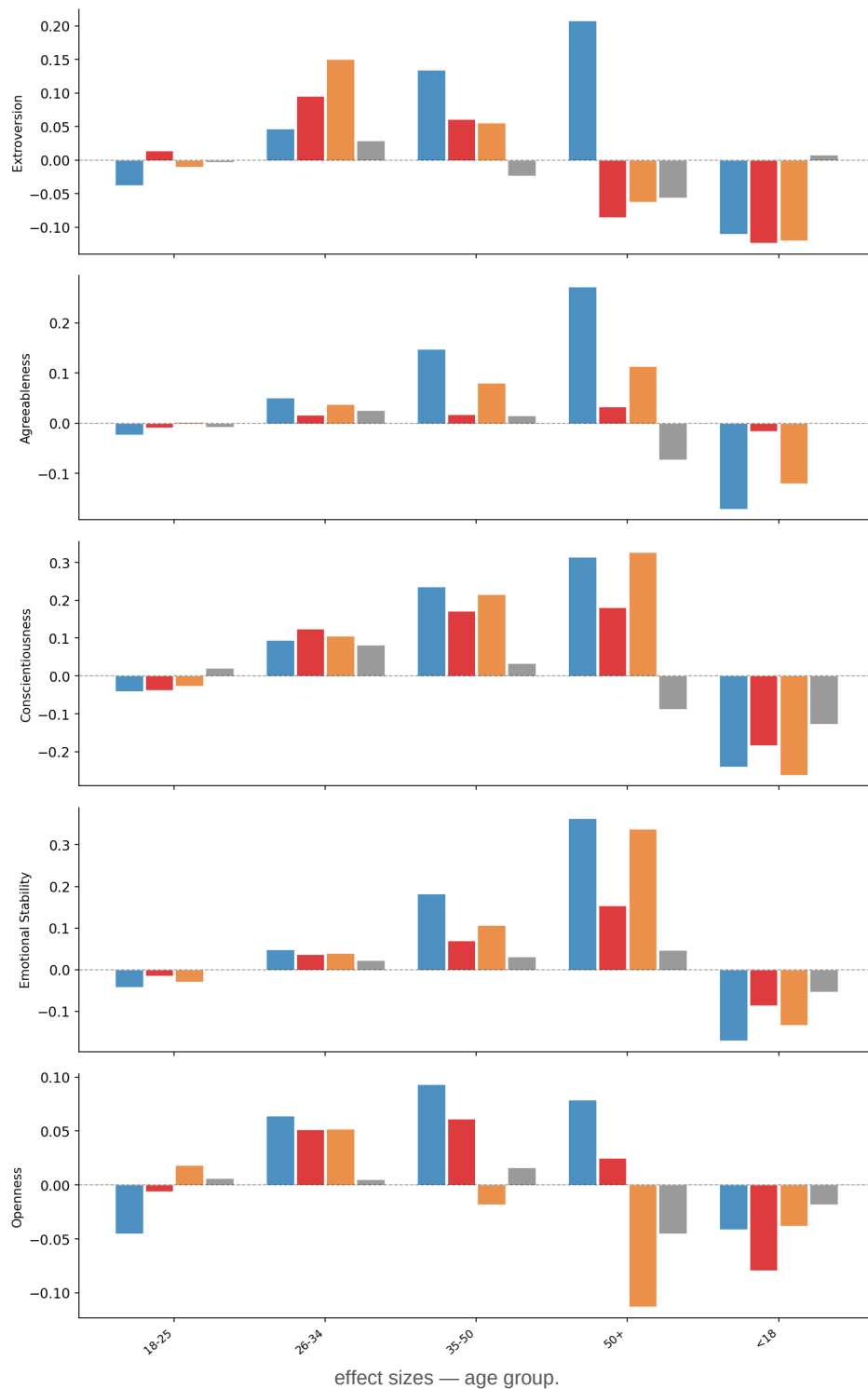
- significant source × factor interactions found for most trait × factor combinations
- models and humans agree on the direction of effects more than on magnitude

4. explicit vs. nl prompting



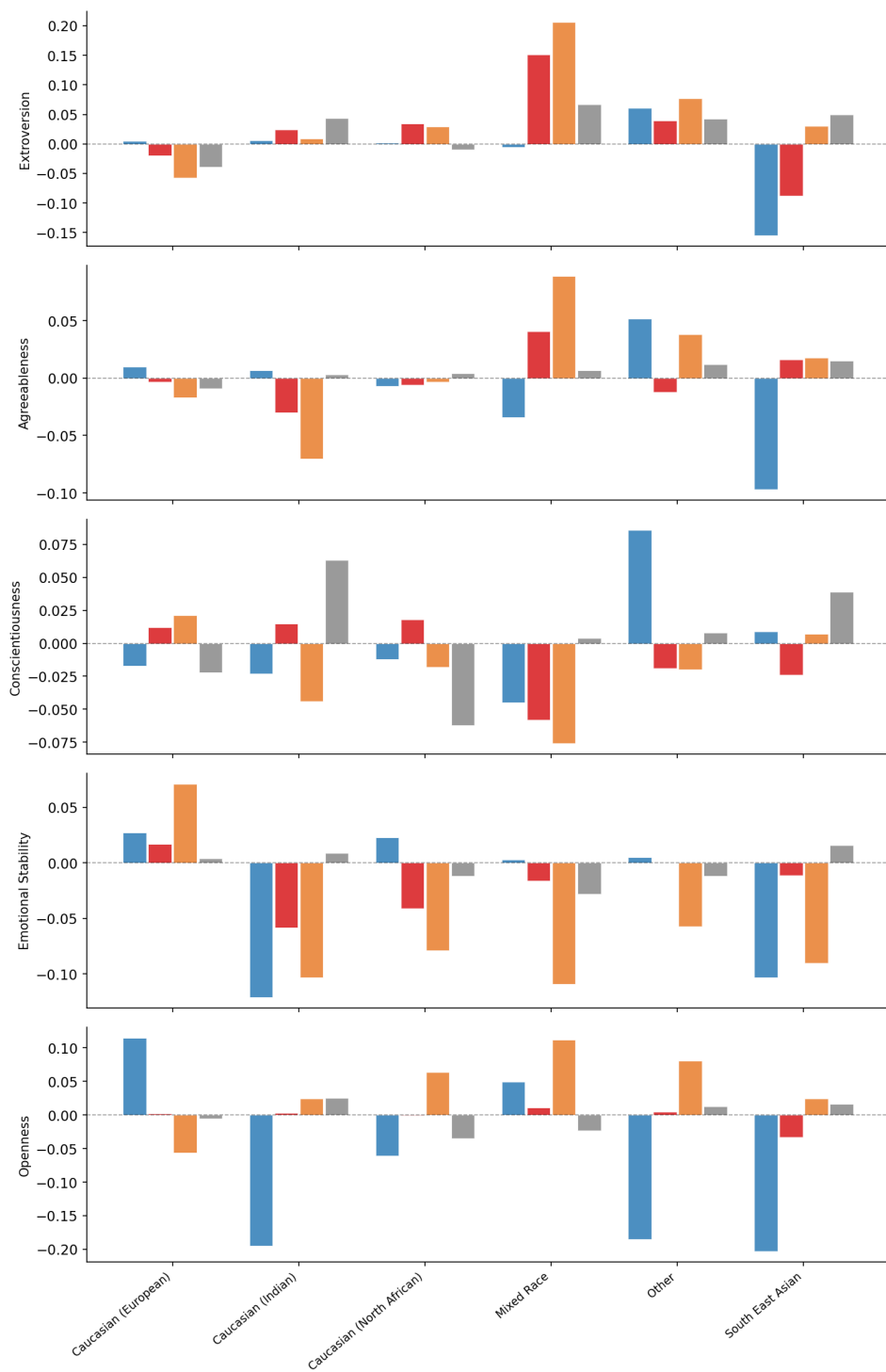
- both gpt conditions preserve the direction of human gender effects on agreeableness and conscientiousness
- gpt explicit systematically compresses extroversion differences; gpt nlp slightly exaggerates them
- emotional stability shows the largest divergence between prompting methods

Demographic Effect Size by Prompting Method (Factor: Age group)
 (bars = subgroup mean – global mean; Qwen NLP note: n=1,200 vs ~20k for others)



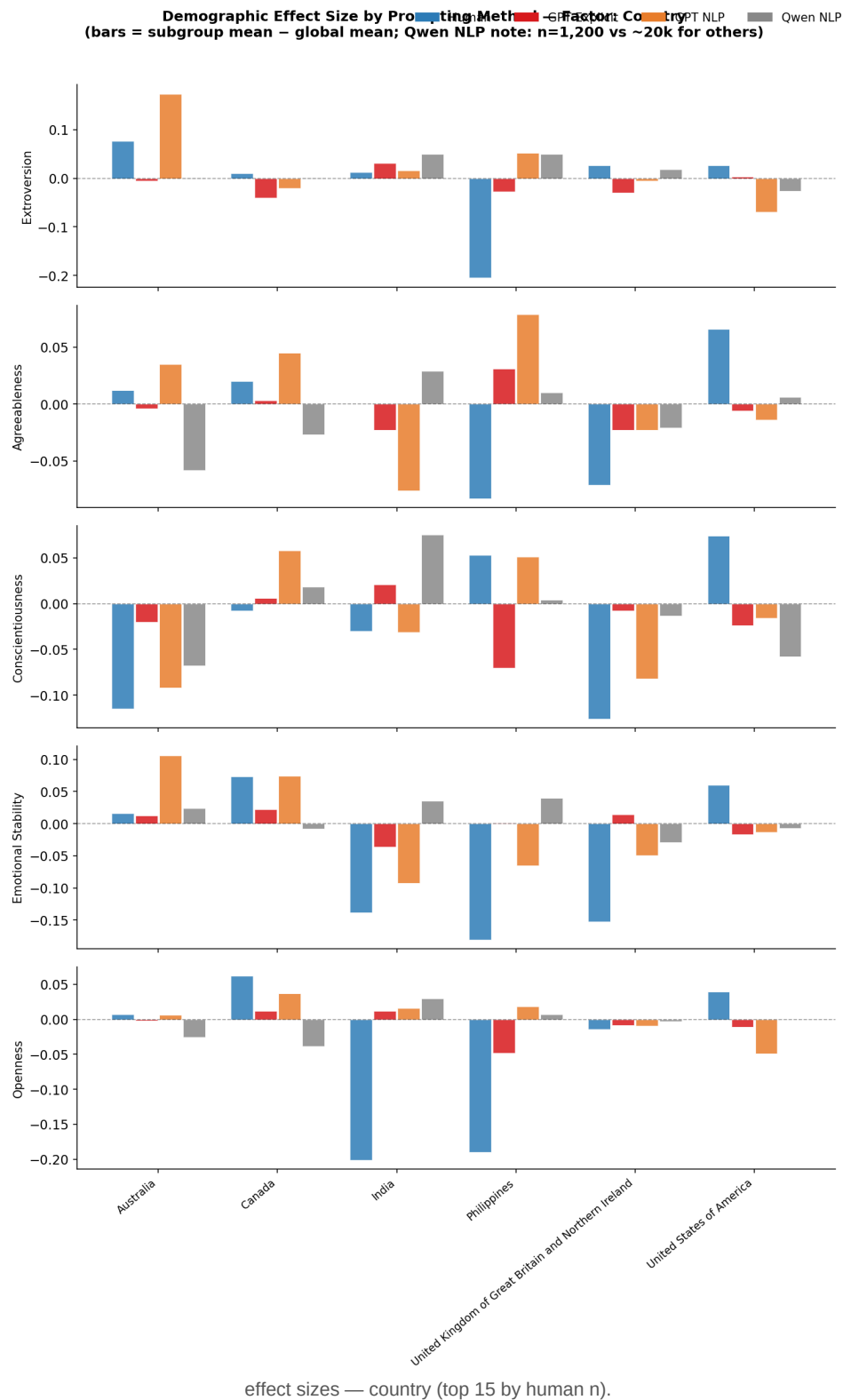
- conscientiousness age gradient (rising with age) is captured by both conditions
- gpt nlp better tracks the human openness decline in older age groups
- emotional stability diverges substantially — both models show older = more stable, but the effect size is larger than in humans

Demographic Effect Size by Prompting Method (bars = subgroup mean – global mean; Qwen NLP note: n=1,200 vs ~20k for others)



effect sizes — race.

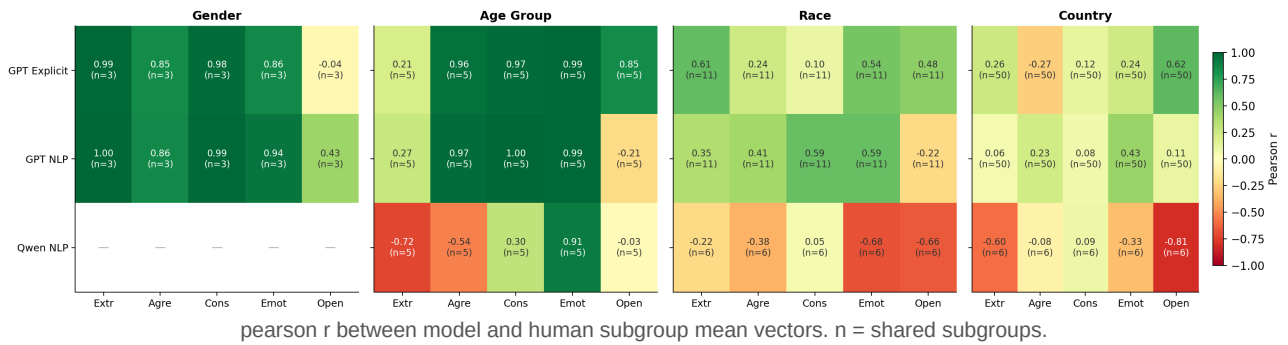
- race effects are most inconsistent — neither model reliably tracks the human pattern
- gpt explicit exaggerates several race effects that are near-zero in humans
- gpt nlp shows some improvement but still diverges for openness and conscientiousness



- country-level effects are the weakest signal in both human and model data
- models do not reliably reproduce the small but real cross-national differences in human data
- overall: gpt nlp is closer to human patterns in 53% of subgroups vs gpt explicit
- gpt explicit exaggerates demographic effects in 23% of subgroups; gpt nlp in 35%
- neither prompting method consistently aligns with humans — nlp is marginally better

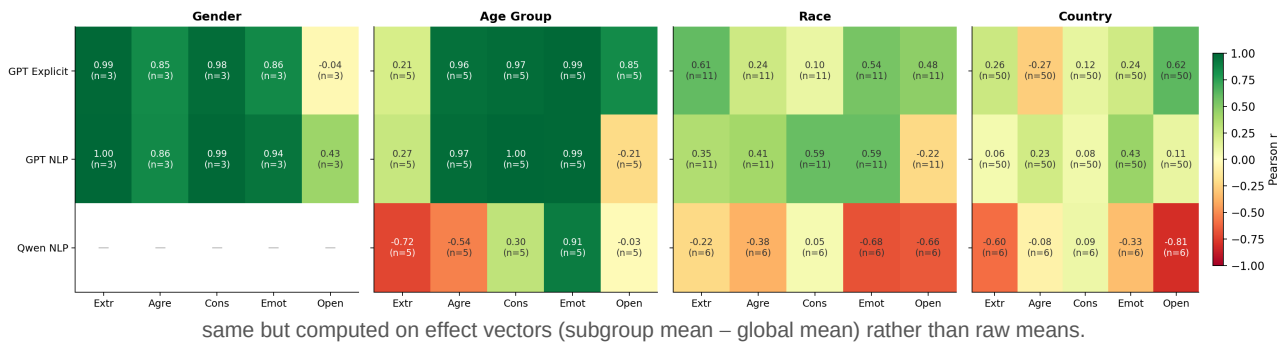
5. alignment with human patterns (pearson r on subgroup means)

Alignment: Model Subgroup Means vs Human Subgroup Means (Pearson r)
n = number of shared subgroups; Qwen NLP has fewer subgroups due to smaller dataset



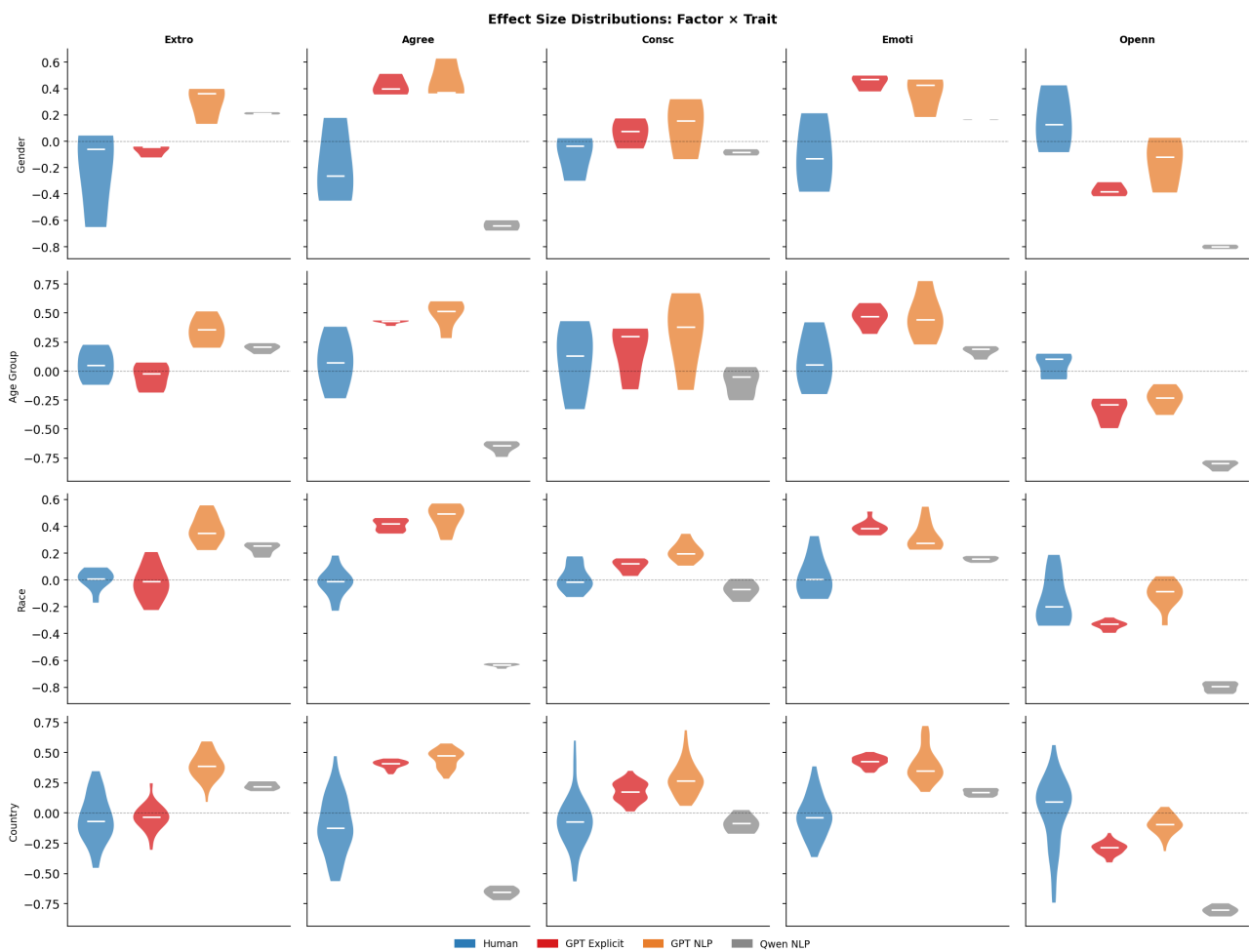
- gpt explicit: mean $r = 0.527$ across all factor×trait combinations
- gpt nlp: mean $r = 0.493$
- best alignment: gpt nlp × gender ($r=0.84$) and gpt explicit × age group ($r=0.80$)
- weakest: both models × country ($r \approx 0.18-0.19$) and race ($r \approx 0.34-0.39$)
- caveat: gender correlation uses only 3 subgroups — confidence intervals are wide

Alignment: Model Demographic Effect Vectors vs Human (Pearson r)
Effect = subgroup_mean – global_mean



- effect-vector correlations are identical to mean correlations — as expected since subtracting a constant doesn't change r
- confirms models capture ordinal ranking of groups better than absolute magnitudes

6. effect size distribution



violin plots of cohen's d (vs human global baseline) per factor × trait × source.

- human cohen's d distribution is narrow and centered near 0 — most subgroups don't differ much from the global mean
- gpt explicit and nlp both show wider tails — more extreme subgroup deviations
- gpt nlp consistently produces larger median effect sizes than explicit across most traits
- race and country factors show the widest spread for gpt conditions

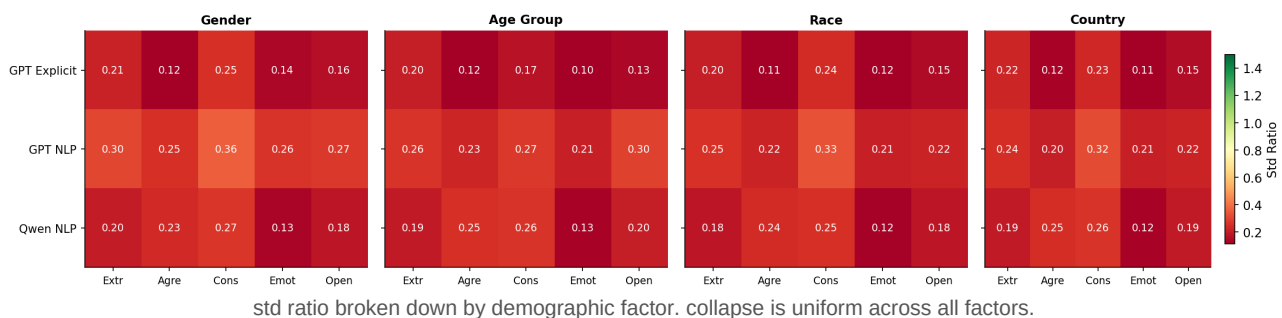
7. variance collapse

Variance Collapse: Mean (Model Within-Subgroup Std / Human Within-Subgroup Std)
< 1 = model produces less within-group diversity; averaged across all factors & subgroups



- gpt explicit: mean std ratio = 0.166 — model diversity is ~17% of human diversity
- gpt nlp: mean std ratio = 0.242 — slightly better but still severe collapse
- verified directly: gpt explicit female extroversion has only 15 unique values (range 2.3–3.7) vs 41 human values (range 1.0–5.0)
- models output near-identical scores for every individual with the same demographic profile — prototype stereotyping at the response level
- this is consistent across all traits and both prompting conditions

Variance Collapse by Factor (Std Ratio model / human)



- no factor escapes the collapse — gender, age, race, and country all show the same pattern
- gpt nlp slightly higher ratios than explicit (0.24 vs 0.17) but the difference is small
- important implication: llm personality simulations should not be used to model individual-level variation

8. caveats & next steps

- qwen nlp results excluded — $n=1,200$ is too small for reliable subgroup-level comparisons
- no qwen explicit condition yet — explicit vs nlp comparison is gpt-4o only
- gender alignment correlation ($r=0.84$) is based on 3 subgroups — statistically unreliable
- country alignment is weak partly due to limited shared subgroups ($n \geq 30$ threshold)
- planned: deepseek-r1-distill-qwen-7b and deepseek-r1-distill-llama-8b (explicit + nlp conditions)
- planned: literature-backed hypothesis testing (3–5 claims from prior work) — colleague handling